# Early Palaeozoic geomagnetism of Gotland

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The majority of the reef limestones and associated lithologies of Gotland are weakly magnetised and contain a complex natural remanent magnetisation, but it is established that they contain an Early Palaeozoic, probably Late Silurian, magnetisation. The Earth's magnetic field was predominantly of normal polarity, with the north pole situated near 170°E and 20°N, but it has not been possible to distinguish any occurrence of a field reversal. In addition, a few rocks show anomalous directions, which are interpreted – in conjunction with some Ordovician results – as indicating the presence in Wenlockian times of a steeply inclined NNW–SSE axis of magnetisation. □ *Palaeomagnetism, pole positions, axis of magnetisation, limestone, Ordovician, Silurian, Lower Palaeozoic, Gotland, south-eastern Sweden, N5655 N5800 E1925 E1805.*

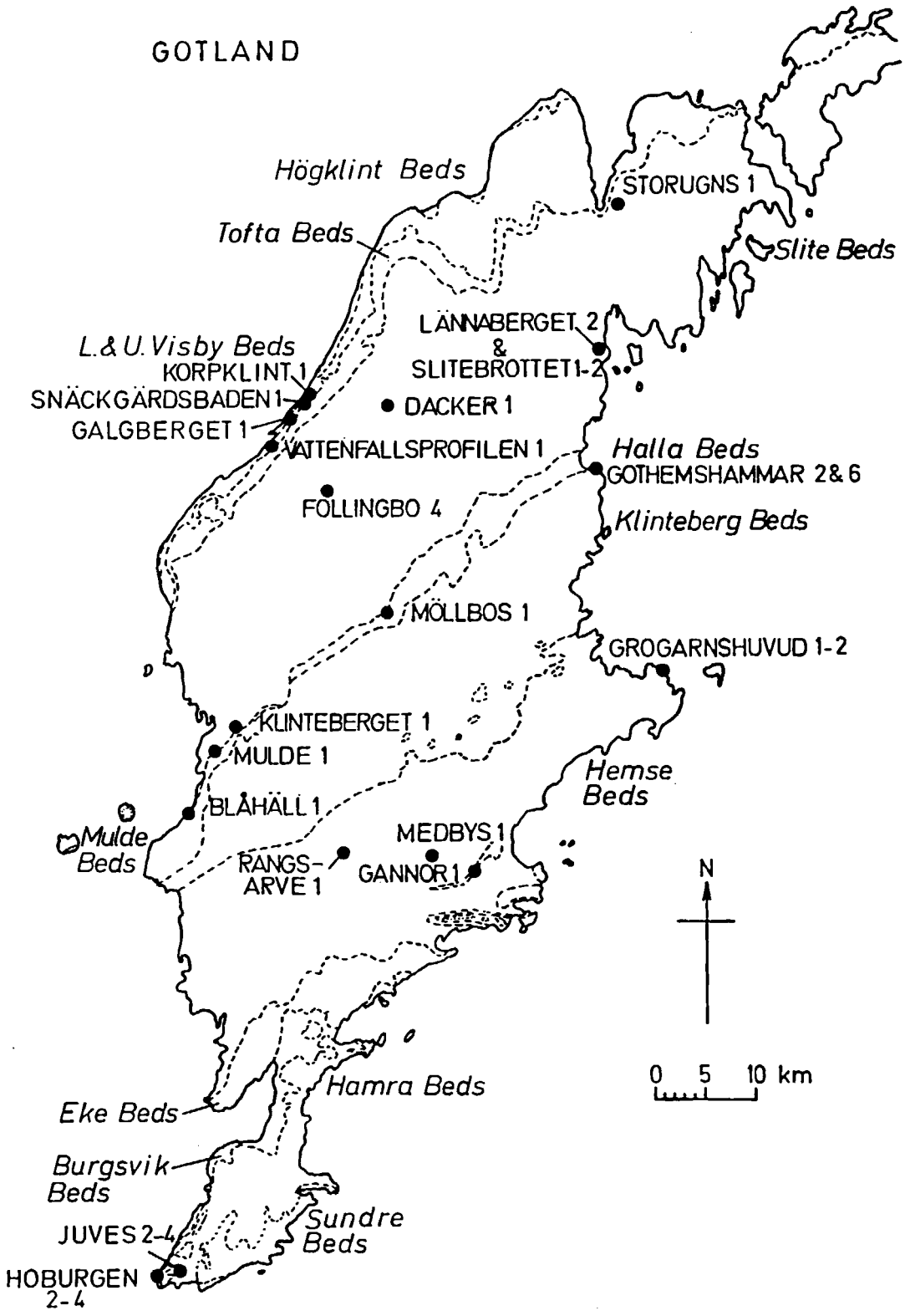
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In the Baltic Syneclise, the gently dipping Precambrian basement is covered by a Lower Palaeozoic sedimentary sequence. Short descriptions of the richly fossiliferous rocks on Gotland have been given by Hede (1960) and Laufeld (1974*a*). The sea withdrew towards the south-east–due to sea-level changes and material infilling – so that the sequence grew horizontally rather than vertically, resulting in lateral differences in lithology and fauna which complicate the correlation of strata. Therefore, it is of interest to investigate whether any reversals of the geomagnetic field have been recorded, since this could define an isochron. Furthermore, as there is some confusion about the Early Palaeozoic pole positions for north-western Europe (Briden et al. 1973), these relatively undisturbed sediments seemed to be a suitable object for a palaeomagnetic investigation.

## Methods of investigation

The present study comprises the results from 84 sites in different lithological and stratigraphical units (Fig. 1). Each site is represented by six cores drilled within an area of less than 0.1 m². One specimen was prepared from each core. A wide variety of rock types, from reef limestones and surrounding facies, were sampled in order to determine which would be suitable for future palaeomagnetic work. With a few exceptions, the rocks show magnetic intensities below $0.1 \times 10^{-6}$ G (numerical data are listed in Claesson 1977) and would ideally need a more sensitive measuring instrument than the Digico spinner magnetometer (Molyneux 1971) that was available. However, the "noise level" of this particular machine, i.e. the signal observed by spinning the holder with no sample in it, remained below $0.020 \times 10^{-6}$ G if measured over $2^8$ revolutions. Although the signal-to-noise ratio is low and no individual results can be relied upon, it is possible, albeit time-consuming, to analyse these rocks. The reader is referred to Tarling (1971) for a survey of palaeomagnetic investigation techniques.

All specimens were measured initially and after partial demagnetisation at 150°C. A few specimens with somewhat higher intensities could be heated to 250 or 300°C. The results are presented in Fig. 2. A self-controlling feed-back system (built by L. Molyneux, School of Physics, Newcastle upon Tyne) for the field-compensation coils of the furnace (Stephenson 1967) kept the residual field below 10 γ, so there is little risk of the directional changes being due to the thermal treatment. The more weakly magnetised specimens were measured several times and the mean specimen result was accepted if the intensity was higher than $0.020 \times 10^{-6}$ G and $\alpha_{95}$ was smaller than 90°. The $\alpha_{95}$ is the radius of the cone of confidence, using a 95% confidence level (Fisher 1953). A site mean (still requiring that $\alpha_{95} < 90°$) was calculated, using at least three accepted specimen-mean results. In most cases, especially after some demagnetisation, the $\alpha_{95}$ of either mean is less than 30°.


GOTLAND
Högklint Beds
STORUGNS 1
Tofta Beds
Slite Beds
LÄNNABERGET 2
&
SLITEBROTTET 1-2
L.& U. Visby Beds
KORPKLINT 1
SNÄCKGÄRDSBADEN 1
DACKER 1
GALGBERGET 1
VATTENFALLSPROFILEN 1
Halla Beds
GOTHEMSHAMMAR 2& 6
FÖLLINGBO 4
Klinteberg Beds
MÖLLBOS 1
GROGARNSHUVUD 1-2
KLINTEBERGET 1
MULDE 1
Hemse Beds
BLÅHÄLL 1
Mulde Beds
MEDBYS 1
RANGS-
ARVE 1
GANNOR 1
N
0 5 10 km
Hamra Beds
Eke Beds
Burgsvik Beds
Sundre Beds
JUVES 2-4
HOBURGEN 2-4

## The predominant, north-easterly direction of magnetisation

Each direction is associated with a degree of uncertainty, so that only statistical behaviour can be considered. The few directions pointing to the SW in Fig. 2 cannot be interpreted as indicators of a reversal; indeed, the direction at 150°C does not refer to the same site as the untreated one. The directions of magnetisation are initially scattered in the north-eastern quadrant and tend to split into two groups after thermal treatment, neither of which is associated with any particular stratigraphical level. The first group moves towards a negative north-easterly direction, and the second clusters around that of the present geomagnetic field. To decide whether this represents a remagnetisation in the present Earth's field, and not an intermediate stage in a move towards, for example, an anti-parallel, south-westerly direction, we need more material with sufficient magnetic intensity to permit the necessary demagnetisations at higher temperatures. It is hardly worth while to pursue the matter, however, since most rock types in the present collection are weakly magnetised. Rocks from only two sites belonging to this second group could be heated to 250°C (Fig. 2C) and above. On the one hand, five specimens from the youngest unit, the "Hoburg marble", retain their directions while the intensity decreases. On the other, the directions of four specimens from the older Hemse Beds at Rangsarve, 1.3–1.4 m below the reference point (Laufeld 1974*b*), remain in the present Earth's field up to 250°C and then move into the south-western quadrant, accompanied by an increase in intensity. The site mean of this latter locality shifts from declination 246° and inclination +33° at 350°C to declination 251° and inclination +40° at 450°C, and the relative movements of the individual specimens, as indicated by the increase in $\alpha_{95}$ from 17° to 25°, appear to be erratic. Whatever the significance of this south-westerly movement, it may be argued that parts of the sequence contain a viscous component along the present geomagnetic field.

A few limestones of the first group contain a magnetisation of stronger intensity and were used to define the direction of the north-easterly component. These were 24 specimens from two red layers at Dacker (Fig. 3), and six brownish specimens from Medbys and Follingbo. If Ö3f– the one deviating specimen – is excluded for the moment, the directions obtained from the Dacker rocks concentrate around declination 28° and inclination −18° ($\alpha_{95}=2°$) after treatment at 250 or 300°C (Table 1). Above 550–600°C, there is a tendency to move away from the north-easterly direction. It is always difficult to interpret movements after demagnetisations at high temperatures. Consistent movements indicate the presence of a previously subordinate, but more stable, component of magnetisation. At the same time, there is an increased degree of uncertainty, because the signal-to-noise ratio is low and magnetically susceptible material is produced during the heating process, although the intensity decrease indicates that this does not become magnetised (Fig. 3). Even if the movements were due to the presence of a second component, it is believed, in the absence of smearing and explicit trends in the specimen behaviour at lower temperatures, that this component does not greatly influence the main north-easterly direction. In addition, the specimens from Medbys and Follingbo (initial intensities 0.65 and $0.23\times 10^{-6}$ G) show a similar direction after demagnetisation at 350 and 250°C, respectively.

## The age of the magnetisation

The north-easterly direction is present in rocks dating from the Late Llandoverian to the Early Ludlovian. Ö3f and the six specimens from site G37 (Fig. 3) appear anomalous. (G37 was taken 1.6–1.7 m above the base of the cliff at Blåhäll; the specimens taken from the base of the cliff are weakly magnetised, although no macroscopic difference in lithology can be seen.) However, the age of any magnetisation is not necessarily that of the rock. Only by comparison with other results can any conclusions be drawn, since it is less likely that the same physico-chemical conditions have prevailed in other areas and lithologies. The results are listed in Table 1 and the left-hand column shows a NE–SW axis of magnetisation (disregarding the error due to the geographical distance), defined by the directions

*Fig. 1.* Geological map of Gotland (after Laufeld 1974*a*), with sampling localities indicated. All the localities are described by Laufeld (1974*b*) and a detailed list of the positions of the sites can be found in Claesson (1977). Dashed lines denote the extension of the different stratigraphical units.

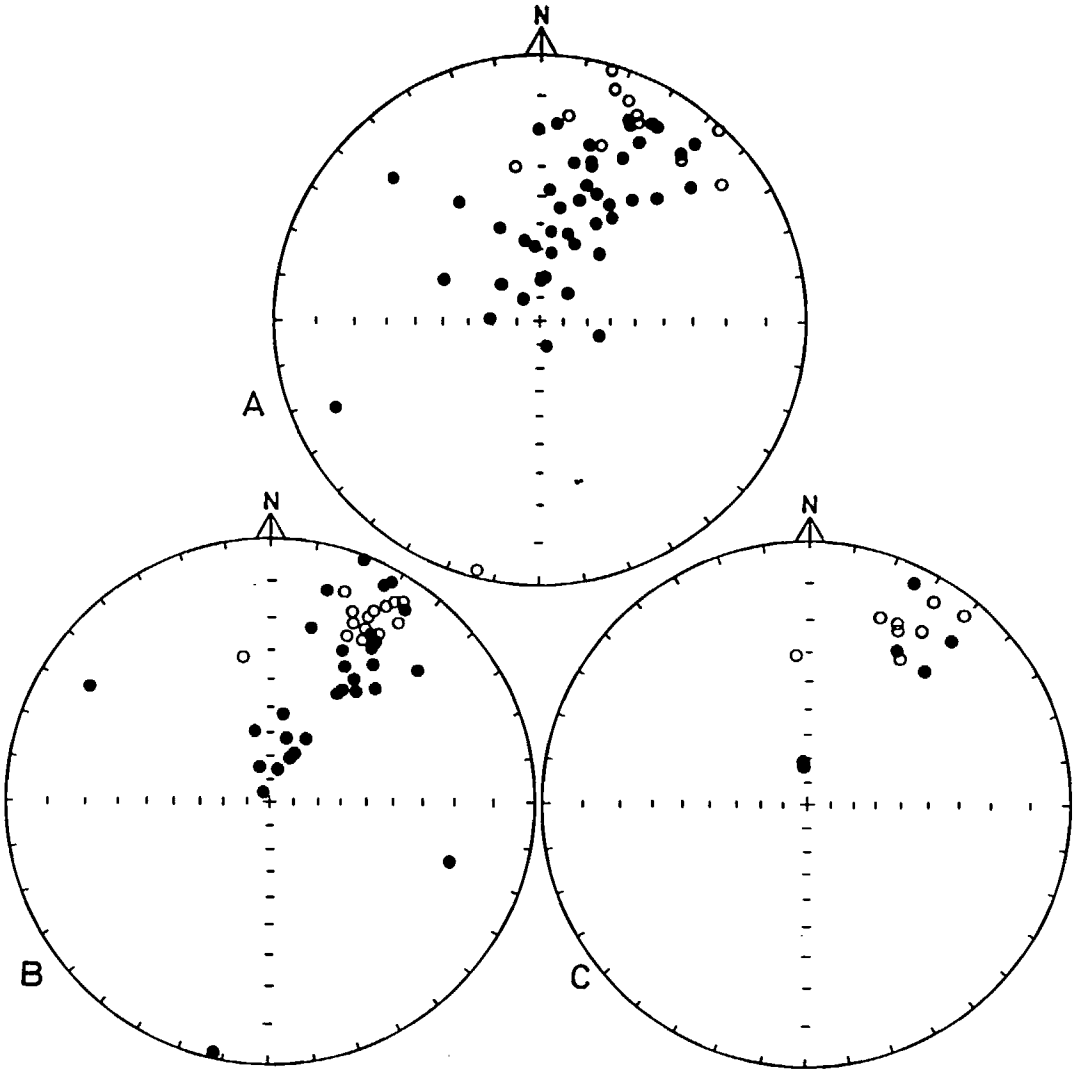



*Fig. 2.* Site mean directions for all accepted results from Gotland. A = initial directions, B = directions after heating to 150°C, and C = after heating to 250°C (300 for some more strongly magnetised rocks). It is important to realise that the collection should be analysed in its entirety and that few individual results can be relied upon. The projections are Wulff nets, where the circumference marks the horizontal and the declination is read clockwise. Dots denote positive (downward) inclination and open circles negative (upward).

of opposite polarity in a reversal system. It must be stressed that most of these values are based on material of poor quality. The most reliable results is that from the Ringerike Sandstone (Norway), where the component was isolated near the Curie temperature of haematite in a few specimens. The pole from the Arrochar Intrusive (Scotland) does not deviate too much from the others and, in view of the thermal behaviour of the contact rocks – which indicates a NE–SW system without defining its end-points (cf. Fig. 3 in Morris et al. 1973) – it is possible that a more detailed demagnetisation of the whole complex would bring it into closer agreement with the Scandinavian position. Other British results (Briden 1970; Morris et al. 1973; Piper 1975) have been critically reviewed elsewhere (Claesson 1977) and the results from eastern Europe are as yet too scanty to allow any comparison to be made. Finally, the Skåne result (Claesson 1977) is based on *one* specimen only, although others from this province show a similar north-easterly movement without reaching any stable point.

*Table 1.* A few Silurian directions of magnetisation and corresponding pole positions from Scandinavia and Great Britain. MS = Middle Silurian, LS = Late Silurian, LS/ED = boundary between Late Silurian and Early Devonian. *D* = declination, read clockwise from north; *I* = inclination, positive if pointing down.

| Rock | Age | °*D* | °*I* | $\alpha_{95}$ | °E | °N | Observations |
|---|---|---|---|---|---|---|---|
| Medbys | LS | 25 | −14 | 8 | 171 | 23 | Based on 6 specimens |
| Dacker (excl. Ö3f) | MS | 28 | −18 | 2 | 169 | 19 | Based on 23 specimens |
| Follingbo | MS | 32 | −13 | 6 | 164 | 21 | Based on 6 specimens |
| Skåne, Southern Sweden | LS | 30 | −15 | 6 | 161 | 22 | Based on 9 heatings between 300 and 500°C (Claesson 1977) |
| Ringerike, Norway | LS | 31 | −11 | | 159 | 21 | Storetvedt et al. (1968) |
| Arrochar, Scotland | LS/ED | 213 | +37 | | 144 | 8 | Briden (1970) |

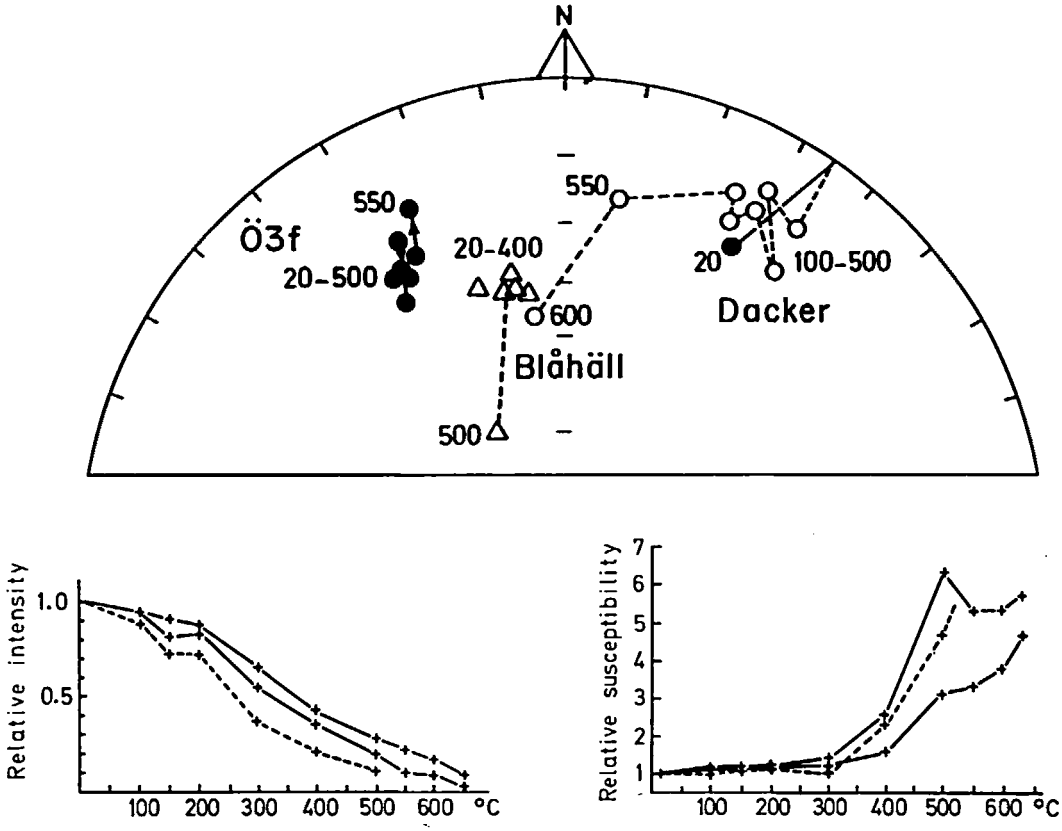



*Fig. 3.* Partial thermal demagnetisation of some of the more strongly magnetised rocks from Dacker (two red layers) and Blåhäll (1.6–1.7 m above base of cliff; grey argillaceous limestone) with demagnetisation temperatures indicated. The negative north-easterly direction is representative of the behaviour of the Dacker rocks, except for the single specimen Ö3f with its positive north-westerly direction. There is no other discernible difference between them. Initial magnitude of site mean magnetisation 0.8–2.4× $10^{-6}$ G, and of susceptibility 1.5–3.1× $10^{-6}$ G/Oe. Corresponding values for Blåhäll are 1.3× $10^{-6}$ G and 5.6× $10^{-6}$ G/Oe. Denotations as in Fig. 2. The susceptibility was measured as described by Collinson et al. (1963).

In contrast to these younger Silurian rocks, those of earliest Silurian and Ordovician age display a different pattern (Claesson 1977, 1978), which is summarised in Fig. 4. This is interpreted as a switch to a NNW–SSE, probably rather steeply inclined, axis of magnetisation, although its *exact* position(s) and appearance cannot be determined on the basis of these data. A similar axis has been reported from British rocks by Piper (1978). The exceptional directions

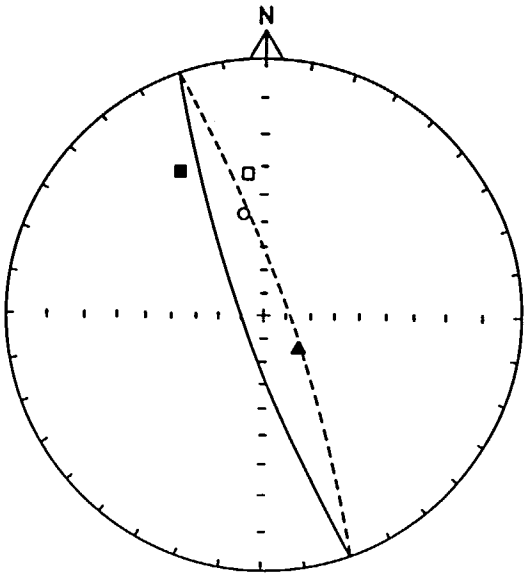



*Fig. 4.* ▲ Mean direction after demagnetisation at 300°C (150°C) or, alternatively, at 150 Oe of Lower Ordovician "Orthoceratite Limestone" from south-eastern Sweden. During further demagnetisation, there is also a movement to the SE, though no second component can be isolated. The great circle is an approximate representation of this movement. A similar pattern appears in a few specimens of Early Silurian age from Skåne in southernmost Sweden and in the "Orthoceratite Limestone" from south-western Sweden. O One site of Middle Ordovician age at Hällekis Quarry in south-western Sweden; ■ Ö3f, mean of results obtained during demagnetisation of a single specimen; □ G37 mean direction after demagnetisation at 250°C. Denotations as in Fig. 2.

of Ö3f and G37 may be taken to indicate that the NNW–SSE axis prevailed into Wenlockian times. Thus, the majority of Gotland rocks would carry a magnetisation which is younger than the rock itself. It is possible to envisage a long period of acquisition during which chemical changes resulting in the development of magnetic minerals took place. The accuracy of the magnetic record depends on the stability of the field *versus* the duration of the magnetisation process. For example, as the case may be here, a few short reversals during a predominantly one-polarity field would only result in a strung distribution of directions and prevent an accurate measurement of the reversal scale. Storetvedt (1967) has suggested that the position of Europe relative to the pole (as in Table 1) remained stable until the Late Carboniferous. Since there is no positive evidence that these Gotland rocks have been remagnetised, I assume that they carry a Late Silurian magnetisation.

## The Grötlingbo core

In addition to the rocks treated above, 187 Silurian and 19 Ordovician specimens of the Swedish Geological Survey core from Grötlingbo on south-eastern Gotland were measured. The measurements were performed on a modified Digico spinner magnetometer, as the samples are about 10 cm high and 6.5 cm in diameter. The core was not orientated and the various layers probably rotated relative to one another during the drilling. Theoretically, it is possible to distinguish a reversal of the geomagnetic field on the basis of inclination only. This does not apply when the inclination of the axis of magnetisation is shallow. One must also remember that variations in inclination may be due to a multi-component magnetisation acquired over an extended period of time, during which the ambient field direction has changed. It is therefore of great importance to ensure that a single component has been isolated. For this reason, the effort was abandoned after testing the behaviour during demagnetisation of a few specimens. Heating to 170°C – a very slow process, in order to ensure that the whole specimen was affected – resulted in an increased instability, as indicated by higher values of $\alpha_{95}$ for the specimen mean direction. Alternating field demagnetisation was also tried, without tumbling, but it was difficult to nullify the external field and the samples seemed to acquire an anhysteretic magnetisation.

## Conclusions

In conclusion, the majority of these samples of reef limestones and associated lithologies are weakly magnetised and any detailed investigation would have to employ a more sensitive magnetometer. Only if the rocks are brown or red in colour can it be assumed, with some degree of certainty, that they carry a magnetisation of which the initial intensity is greater than $0.1 \times 10^{-6}$ G. These rocks contain a larger amount of magnetic minerals, mostly as red haematite-pigment, although the reason for this increase has not been investigated. It has been established, however, that they contain an Early Palaeozic, probably Late Silurian, magnetisation. The north pole was situated near 170°E and 20°N, and it has not been possible to distinguish any occurrence of a field reversal. Reversals of the geomagnetic field may not have taken place during this period of time or, alternatively,

brief polarity changes may have been obscured, due to the extended magnetisation process. In either case, the palaeomagnetic method of investigation is not, at this stage, able to provide a simple and rapid tool for dating and/or correlation of the Gotland rocks.

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A contribution to
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## References

Briden, J. C., 1970: Palaeomagnetic results from the Arrochar and Garabal Hill–Glen Fyne igneous complexes, Scotland. *Geophysical Journal of the Royal Astronomical Society 21*, 457–470.

Briden, J. C., Morris W. A. & Piper, J. D. A., 1973: Palaeomagnetic studies in the British Caledonides: VI. Regional and global implications. *Geophysical Journal of the Royal Astronomical Society 34*, 107–134.

Claesson, K. C., 1977: *Palaeomagnetic studies of Swedish Lower Palaeozoic rocks*. 349 pp. University of Newcastle upon Tyne. [Unpublished Ph.D. thesis.]

Claesson, C., 1978: Swedish Ordovician limestones: problems in clarifying their directions of magnetization. *Physics of the Earth and Planetary Interiors 16*, 65–72.

Collinson, D. W., Molyneux, L. & Stone, D. B., 1963: A total and anisotropic magnetic susceptibility meter. *Journal of Scientific Instruments 40*, 310–312.

Fisher, R., 1953: Dispersion on a sphere. *Proceedings of the Royal Society of London A 217*, 295–305.

Hede, J. A., 1960: The Silurian of Gotland. *In* G. Regnéll & J. A. Hede: The Lower Palaeozoic of Scania. The Silurian of Gotland. *International Geological Congress, XXI Session, Norden*, 44–89. Stockholm.

Laufeld, S., 1974*a*: Silurian Chitinozoa from Gotland. *Fossils and Strata 5*, 1–130.

Laufeld, S., 1974b: Reference localities for palaeontology and geology in the Silurian of Gotland. *Sveriges geologiska undersökning C 705*, 1–172.

Molyneux, L., 1971: A complete result magnetometer for measuring the remanent magnetization of rocks. *Geophysical Journal of the Royal Astronomical Society 24*, 429–433.

Morris, W. A., Briden, J. C., Piper, J. D. A. & Sallomy, J. T., 1973: Palaeomagnetic studies in the British Caledonides: V. Miscellaneous new data. *Geophysical Journal of the Royal Astronomical Society 34*, 69–105.

Piper, J. D. A., 1975: Palaeomagnetism of Silurian lavas of Somerset and Gloucestershire, England. *Earth and Planetary Science Letters 25*, 355–360.

Piper, J. D. A., 1978: Palaeomagnetic survey of the (Palaeozoic) Shelve inlier and Berwyn Hills, Welsh Borderlands. *Geophysical Journal of the Royal Astronomical Society 53*, 355–371.

Stephenson, A., 1967: Apparatus for thermal demagnetization by the progressive method. *In* D. W. Collinson, K. M. Creer & S. K. Runcorn (eds.): *Methods in palaeomagnetism*, 296–298. Elsevier, Amsterdam.

Storetvedt, K. M., 1967: A synthesis of the Palaeozoic palaeomagnetic data for Europe. *Earth and Planetary Science Letters 3*, 444–448.

Storetvedt, K. M., Halvorsen, E. & Gjellestad, G., 1968: Thermal analysis of the natural remanent magnetism of some Upper Silurian red sandstones in the Oslo region. *Tectonophysics 5*, 413–426.

Tarling, D. H., 1971: *Principles and applications of palaeomagnetism*. 164 pp. Chapman and Hall, London.